

Problem F. Simple Calculations

Input file: `calc.in`
Output file: `calc.out`
Time limit: 0.25 seconds

You have to calculate the function $F(n, k) = (2^{2^n} + 1) \bmod k$.

Input

Input file contains two integer numbers n and k separated by spaces ($0 \leq n \leq 2^{31} - 1$, $1 \leq k \leq 10^6$).

Output

You must output $F(n, k)$.

Example

<code>calc.in</code>	<code>calc.out</code>
0 4	3